

SCORE A BOWLING GAME!

Introduction

This coding challenge will task you to create a Python application/library that will score a bowling game. **No UI is required.** Focus on clean scoring logic, input validation, and automated tests. You may structure the project however you like, as long as it can be run and tested easily.

Coding Considerations and Work Submission

You are free to program this application however you wish. Treat this code as if you were programming **production code for a product**.

The code should be tested and work as expected.

When completed, upload to a repository (Github, etc.) and **share the link** to the project.

Input/Output

Input

A bowling game consists of **10 frames**.

Provide your scorer one of the following input styles (choose one and document it in your README):

Option A: Frames

- Frames 1-9: each frame has **up to 2 rolls**
- Frame 10: **has 2 or 3 rolls** depending on strike/spare

Example:

```
[ ["8", "/"], ["5", "4"], ["9", "0"], ["X"], ["X"], ["5", "/"], ["5", "3"], ["6", "3"], ["9", "/"], ["9", "/", "X"] ]
```

Option B: Rolls

- A flat list of rolls in order

Example:

```
["8", "/", "5", "4", "9", "0", "X", "X", "5", "/", "5", "3", "6", "3", "9", "/", "9", "/", "X"]
```

Output

Return the **cumulative score at the end of each frame** (10 values).

If you choose to support partially complete games, frames that cannot yet be scored may return None (document your approach).

Valid Input Values

- an "x" or an "X" for a value for a **STRIKE**
- an "/" for a **spare**
- 0-9 for each shot if it is not a strike or a spare

Logic

* Strike - if you knock down all 10 pins in the first shot of a frame, this is a strike. A strike is worth 10 points PLUS THE SUM OF YOUR NEXT TWO SHOTS.

* Spare - If you knock down all 10 pins using both shots of a frame, this is a spare. A spare is worth 10 points PLUS THE SUM OF YOUR NEXT ONE SHOT.

* Open frame - If you do not knock down all 10 pins using both shots of your frame (9 or fewer pins knocked down), this is an open frame and is worth the number of pins knocked down.

* TENTH FRAME - Strike on first roll → two additional rolls, spare in first two rolls → one additional roll, open frame after two rolls → game ends, the 10th frame score is simply the total pins knocked down in the 10th frame (including bonus rolls)

EXAMPLE GAME

1	2	3	4	5	6	7	8	9	10
8 /	5 4	9 0	X _	X _	5 /	5 3	6 3	9 /	9 / X
15	24	33	58	78	93	101	110	129	149

Frame 1 - a spare which is 10 plus the number of pins knocked over in the next shot which was 5, so 15 points.

Frame 2 - Open frame of 9 points, so add 9 to the previous score of 15 for 24.

Frame 3 - Open frame of 9 points, so add 9 to the previous score of 24 for 33.

Frame 4 - Strike - earn 10 points plus the sum of the next two shots. In this case, the next two shots are a strike and a 5 which totals 15 pins, so $10 + 15 = 25$ points earned for Frame 4 for a total of 58.

Frame 5 - Strike - earn 10 points plus the sum of the next two shots. In this case, the next two shots are a 5 and a spare (a 5) which is a total of 10 pins, so $10 + 10 = 20$ and added to 58 gives a score of 78.

Frame 6 - a spare which is 10 plus the number of pins knocked over in the next shot which is 5 so 15 points added gives a score of 93.

Frame 7 - Open frame of 8 points, so add 8 to the previous score of 93 for 101.

Frame 8 - Open frame of 9 points, so add 9 to the previous score of 101 for 110.

Frame 9 - a spare which is 10 plus the number of pins knocked over in the next shot, which is a 9, so 10+9 is 19 plus a previous score of 110 gives the score 129.

Frame 10 - Last frame. A spare in the first two shots gives an extra roll, which is a strike. The score in the 10th frame is simply the number of pins knocked down total, which is 20, giving a final score of 149.

Validation Requirements

Your implementation should validate input and raise a clear exception (e.g., `ValueError`) for invalid games. At minimum, handle:

- Spare ("/") cannot appear as the first roll of a frame
- Frame totals cannot exceed 10 unless the second roll is a spare indicator
- Tenth-frame extra rolls only allowed when earned (strike/spare)
- No extra rolls beyond a complete game
- Invalid symbols (anything other than X/x, /, 0–9)

Document any additional assumptions.

Testing Requirements (pytest)

Your submission **must include automated tests**. Use pytest (preferred) or another common Python test framework and document how to run it.

Minimum required test cases

1. The provided **example game** returns:
 - [15, 24, 33, 58, 78, 93, 101, 110, 129, 149]
2. **Perfect game** (12 strikes) returns 300
3. **All spares** (e.g., 5/ in frames 1–10 with a 5 bonus) returns 150
4. **All open frames** (no strikes/spares) returns the correct total
5. Tenth frame behavior:
 - strike + two bonus rolls
 - spare + one bonus roll
 - open frame ends the game (no bonus)
6. Validation tests (at least 5):
 - spare in first roll

- invalid character
- too many rolls in 10th
- frame pin count exceeds 10 without spare
- extra rolls after game completion

Nice-to-have

- Parameterized tests for multiple scenarios
- Tests for partial scoring (if supported)

Thank you!